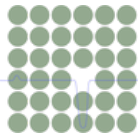




UNIVERSITÄT
LEIPZIG



Edgar Harutyunyan

Introduction

November 28, 2025

Edgar Harutyunyan

Molecular Nanophotonics Group
Peter Debye Institute for Soft Matter Physics
Universität Leipzig

Education

- **Bachelor of Science in International Physics Studies Program (Honours)**
 - Leipzig University, Leipzig, Germany
 - Oct 2022 – Sep 2026
- **Bachelor of Informatics in Information Technologies**
 - National Polytechnic University of Armenia, Gyumri, Armenia
 - Sep 2016 – Jul 2020

Experience

– Student Assistant

- Faculty of Physics and Earth System Sciences, Leipzig University
- Oct 2024 – Current

– Software Engineer

- Instigate Design CJSC, Yerevan, Armenia
- Oct 2021 – Jan 2025 (Remote since Feb 2023)
- **Key Projects:**
 - **Heterogeneous Computing & AI:** Text semantic analysis on FPGA/CPU hybrid system.
 - **Satellite Systems:** High-throughput chip-to-chip data transfer & FPGA-based UHR image analysis.
 - **Kernel Development:** Enhanced multi-threading library for RISC-V.
 - **Custom Processor Design:** Designed custom CPU architectures and instruction sets.

Skills

Programming & Software

- C/C++ (STL, OOP, Gcc/Gdb, Parallel Programming)
- Python (OOP, ML: Pandas/Scikit-learn, Data Analysis: Numpy/Matplotlib)
- MATLAB
- Linux, Bash, GNU Make, Vim
- Version Control (Git, SVN)

Hardware & Low-Level

- Hardware Design (Verilog/VHDL, Xilinx Vivado/Vitis)
- Low-Level Programming (x86/ARM, MIPS, RISC-V)

Soft Skills

- Team Collaboration
- Agile Development
- Planning & Estimation

Languages & Certifications

Languages

- **Armenian:** Native
- **English:** Professional
- **Russian:** Professional
- **German:** Learning actively

Certifications

- FPGA Conference 2024 (Munich)
- Machine Learning (Armenian Code Academy)
- Quality Engineering (Business Development Group)